CS173: Discrete Structures
Problem Set 5

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1. We say that a set $\boldsymbol{x}$ is **$\in$-transitive** $:\Leftrightarrow \forall \boldsymbol{y}(\boldsymbol{y} \in \boldsymbol{x} \Rightarrow \forall \boldsymbol{z}(\boldsymbol{z} \in \boldsymbol{y} \Rightarrow \boldsymbol{z} \in \boldsymbol{x}))$.

   Prove that every natural number is $\in$-transitive.

   Proof. We will show $(\forall \boldsymbol{n} \in \mathbb{N})(\boldsymbol{n}$ is $\in$-transitive$)$ by weak induction.

   Basis Step: We must show $\mathbf{0}$ is $\in$-transitive, i.e., $\forall \boldsymbol{y}(\boldsymbol{y} \in \mathbf{0} \Rightarrow \forall \boldsymbol{z}(\boldsymbol{z} \in \boldsymbol{y} \Rightarrow \boldsymbol{z} \in \mathbf{0}))$. Since $\mathbf{0} = \emptyset$, there is no $\boldsymbol{y}$ such that $\boldsymbol{y} \in \mathbf{0}$. Therefore any statement $\forall \boldsymbol{y}(\boldsymbol{y} \in \mathbf{0} \Rightarrow \boldsymbol{p})$ where p is an arbitrary proposition holds.

   Inductive Step: Let $\boldsymbol{k} \in \mathbb{N}$, and assume $\boldsymbol{k}$ is $\in$-transitive. We will show $\mathfrak{suc}(\boldsymbol{k})$ is $\in$-transitive.

   Recall $\mathfrak{suc}(\boldsymbol{k}) = \boldsymbol{k} \cup \{\boldsymbol{k}\}$ by definition. Let $\boldsymbol{y}$ be arbitrary and suppose $\boldsymbol{y} \in \mathfrak{suc}(\boldsymbol{k})$. Then $\boldsymbol{y} \in \boldsymbol{k} \cup \{\boldsymbol{k}\}$, so either $\boldsymbol{y} \in \boldsymbol{k}$ or $\boldsymbol{y} \in \{\boldsymbol{k}\}$, meaning either $\boldsymbol{y} \in \boldsymbol{k}$ or $\boldsymbol{y} = \boldsymbol{k}$.

   Now let $\boldsymbol{z}$ be arbitrary and suppose $\boldsymbol{z} \in \boldsymbol{y}$. We consider two cases.

   Case 1: Suppose $\boldsymbol{y} \in \boldsymbol{k}$. Since $\boldsymbol{k}$ is $\in$-transitive by the inductive hypothesis and $\boldsymbol{y} \in \boldsymbol{k}$ and $\boldsymbol{z} \in \boldsymbol{y}$, we conclude $\boldsymbol{z} \in \boldsymbol{k}$. Since $\boldsymbol{k} \subseteq \boldsymbol{k} \cup \{\boldsymbol{k}\} = \mathfrak{suc}(\boldsymbol{k})$, we have $\boldsymbol{z} \in \mathfrak{suc}(\boldsymbol{k})$.

   Case 2: Suppose $\boldsymbol{y} = \boldsymbol{k}$. Then $\boldsymbol{z} \in \boldsymbol{y} = \boldsymbol{k}$, so $\boldsymbol{z} \in \boldsymbol{k} \subseteq \boldsymbol{k} \cup \{\boldsymbol{k}\} = \mathfrak{suc}(\boldsymbol{k})$, giving us $\boldsymbol{z} \in \mathfrak{suc}(\boldsymbol{k})$.

   In either case, $\boldsymbol{z} \in \mathfrak{suc}(\boldsymbol{k})$. Since $\boldsymbol{y}$ and $\boldsymbol{z}$ were arbitrary, $\mathfrak{suc}(\boldsymbol{k})$ is $\in$-transitive.

   Therefore, we conclude $(\forall \boldsymbol{n} \in \mathbb{N})(\boldsymbol{n}$ is $\in$-transitive$)$.

2. We will work up to a proof of commutativity of addition.

(a) Prove $(\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{N})(\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{y}) = \mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{y})$.

Proof. Let $\boldsymbol{x} \in \mathbb{N}$. We will prove $(\forall \boldsymbol{y} \in \mathbb{N})(\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{y}) = \mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{y})$ by weak induction.

Basis Step: We observe the following.

$$\mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{0} = \mathfrak{suc}(\boldsymbol{x}) \qquad \text{(by definition of } +\text{)}$$

$$\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{0}) = \mathfrak{suc}(\boldsymbol{x}) \qquad \text{(by definition of } +\text{)}$$

Therefore $\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{0}) = \mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{0}$.

Inductive Step: Let $\boldsymbol{k} \in \mathbb{N}$, and assume $\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{k}) = \mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{k}$. We observe the following.

$$\begin{aligned}
\mathfrak{suc}(\boldsymbol{x}) + \mathfrak{suc}(\boldsymbol{k}) &= \mathfrak{suc}(\mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{k}) && \text{by definition of } + \\
&= \mathfrak{suc}(\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{k})) && \text{by the inductive hypothesis} \\
&= \mathfrak{suc}(\boldsymbol{x} + \mathfrak{suc}(\boldsymbol{k})) && \text{by definition of } +
\end{aligned}$$

Therefore $\mathfrak{suc}(\boldsymbol{x} + \mathfrak{suc}(\boldsymbol{k})) = \mathfrak{suc}(\boldsymbol{x}) + \mathfrak{suc}(\boldsymbol{k})$.
We can therefore conclude $(\forall \boldsymbol{y} \in \mathbb{N})(\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{y}) = \mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{y})$, and since $\boldsymbol{x}$ was arbitrary, $(\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{N})(\mathfrak{suc}(\boldsymbol{x}+\boldsymbol{y}) = \mathfrak{suc}(\boldsymbol{x}) + \boldsymbol{y})$.

(b) Prove $(\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{N})(\boldsymbol{x}+\boldsymbol{y} = \boldsymbol{y}+\boldsymbol{x})$.

Proof. Let $\boldsymbol{x} \in \mathbb{N}$. We will prove $(\forall \boldsymbol{y} \in \mathbb{N})(\boldsymbol{x}+\boldsymbol{y} = \boldsymbol{y}+\boldsymbol{x})$ by weak induction.

Basis Step: We observe the following.

$$\begin{aligned}
\boldsymbol{x} + \boldsymbol{0} &= \boldsymbol{x} && \text{by Theorem: Zero is the Additive Identity} \\
&= \boldsymbol{0} + \boldsymbol{x} && \text{by Theorem: Zero is the Additive Identity}
\end{aligned}$$

Therefore $\boldsymbol{x} + \boldsymbol{0} = \boldsymbol{0} + \boldsymbol{x}$ as desired.

Inductive Step: Let $\boldsymbol{k} \in \mathbb{N}$, and assume $\boldsymbol{x}+\boldsymbol{k} = \boldsymbol{k}+\boldsymbol{x}$. We observe the following.

$$\begin{aligned}
\boldsymbol{x} + \mathfrak{suc}(\boldsymbol{k}) &= \mathfrak{suc}(\boldsymbol{x}+\boldsymbol{k}) && \text{by definition of } + \\
&= \mathfrak{suc}(\boldsymbol{k}+\boldsymbol{x}) && \text{by the inductive hypothesis} \\
&= \mathfrak{suc}(\boldsymbol{k}) + \boldsymbol{x} && \text{by part (a)}
\end{aligned}$$

Therefore $\boldsymbol{x} + \mathfrak{suc}(\boldsymbol{k}) = \mathfrak{suc}(\boldsymbol{k}) + \boldsymbol{x}$ as desired.
We can therefore conclude $(\forall \boldsymbol{y} \in \mathbb{N})(\boldsymbol{x}+\boldsymbol{y} = \boldsymbol{y}+\boldsymbol{x})$, and since $\boldsymbol{x}$ was arbitrary, $(\forall \boldsymbol{x}, \boldsymbol{y} \in \mathbb{N})(\boldsymbol{x}+\boldsymbol{y} = \boldsymbol{y}+\boldsymbol{x})$.

3. Show that $(\forall x, y \in \mathbb{N})(x < y \Rightarrow (\exists n \in \mathbb{N})(x + n = y))$.

Proof. Let $x \in \mathbb{N}$. We will prove $(\forall y \in \mathbb{N})(x < y \Rightarrow (\exists n \in \mathbb{N})(x + n = y))$ by weak induction.

Basis Step: We must show $x < 0 \Rightarrow (\exists n \in \mathbb{N})(x + n = 0)$. Recall that $x < 0$ means $x \in 0 = \emptyset$ by definition, which is false for any x. Therefore the implication $x < 0 \Rightarrow \dots$ holds vacuously.

Inductive Step: Let $k \in \mathbb{N}$, and assume $x < k \Rightarrow (\exists n \in \mathbb{N})(x + n = k)$. We must show $x < \mathfrak{suc}(k) \Rightarrow (\exists n \in \mathbb{N})(x + n = \mathfrak{suc}(k))$.

Suppose $x < \mathfrak{suc}(k)$, meaning $x \in \mathfrak{suc}(k) = k \cup \{k\}$. Then either $x \in k$ or $x \in \{k\}$, i.e., either $x < k$ or $x = k$.

Case 1: Suppose $x < k$. By the inductive hypothesis, there exists $n_0 \in \mathbb{N}$ such that $x + n_0 = k$. Then observe:

$$\begin{aligned} x + \mathfrak{suc}(n_0) &= \mathfrak{suc}(x + n_0) && \text{by definition of } + \\ &= \mathfrak{suc}(k) && \text{since } x + n_0 = k \end{aligned}$$

So $\mathfrak{suc}(n_0) \in \mathbb{N}$ witnesses $(\exists n \in \mathbb{N})(x + n = \mathfrak{suc}(k))$.

Case 2: Suppose $x = k$. Then observe:

$$\begin{aligned} x + 1 &= \mathfrak{suc}(x) && \text{by Lemma 6.5} \\ &= \mathfrak{suc}(k) && \text{since } x = k \end{aligned}$$

So $1 \in \mathbb{N}$ witnesses $(\exists n \in \mathbb{N})(x + n = \mathfrak{suc}(k))$.

In either case, $(\exists n \in \mathbb{N})(x + n = \mathfrak{suc}(k))$ holds. We therefore conclude $x < \mathfrak{suc}(k) \Rightarrow (\exists n \in \mathbb{N})(x + n = \mathfrak{suc}(k))$.

Since x was arbitrary, $(\forall x, y \in \mathbb{N})(x < y \Rightarrow (\exists n \in \mathbb{N})(x + n = y))$ as desired.

4. Prove that $(\forall \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{N})(\boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{z}) = (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{z}))$.

Proof. Let $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{N}$. We will prove $(\forall \boldsymbol{z} \in \mathbb{N})(\boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{z}) = (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{z}))$ by weak induction.

Basis Step: We observe the following.

$$\begin{aligned} \boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{0}) &= \boldsymbol{x} \cdot \boldsymbol{y} && \text{by definition of } + \\ &= (\boldsymbol{x} \cdot \boldsymbol{y}) + \boldsymbol{0} && \text{by definition of } \cdot \\ &= (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{0}) && \text{by definition of } \cdot \end{aligned}$$

Therefore $\boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{0}) = (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{0})$ as desired.

Inductive Step: Let $\boldsymbol{k} \in \mathbb{N}$, and assume $\boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{k}) = (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{k})$. We observe the following.

$$\begin{aligned} \boldsymbol{x} \cdot (\boldsymbol{y} + \mathfrak{suc}(\boldsymbol{k})) &= \boldsymbol{x} \cdot \mathfrak{suc}(\boldsymbol{y} + \boldsymbol{k}) && \text{by definition of } + \\ &= \boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{k}) + \boldsymbol{x} && \text{by definition of } \cdot \\ &= ((\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{k})) + \boldsymbol{x} && \text{by the inductive hypothesis} \\ &= (\boldsymbol{x} \cdot \boldsymbol{y}) + ((\boldsymbol{x} \cdot \boldsymbol{k}) + \boldsymbol{x}) && \text{by associativity of } + \\ &= (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \mathfrak{suc}(\boldsymbol{k})) && \text{by definition of } \cdot \end{aligned}$$

Therefore $\boldsymbol{x} \cdot (\boldsymbol{y} + \mathfrak{suc}(\boldsymbol{k})) = (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \mathfrak{suc}(\boldsymbol{k}))$.

We can therefore conclude $(\forall \boldsymbol{z} \in \mathbb{N})(\boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{z}) = (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{z}))$, and since $\boldsymbol{x}$ and $\boldsymbol{y}$ were arbitrary, $(\forall \boldsymbol{x}, \boldsymbol{y}, \boldsymbol{z} \in \mathbb{N})(\boldsymbol{x} \cdot (\boldsymbol{y} + \boldsymbol{z}) = (\boldsymbol{x} \cdot \boldsymbol{y}) + (\boldsymbol{x} \cdot \boldsymbol{z}))$.

5. Given a sequence of natural numbers $\boldsymbol{f} : \mathbb{N} \to \mathbb{N}$ and a natural number $\boldsymbol{a} \in \mathbb{N}$, we recursively define the iterated summation of finitely many values of $\boldsymbol{f}$ as follows:

$$\sum_{i=a}^{a} f(i) := f(a)$$

$$\sum_{i=a}^{\mathfrak{suc}(b)} f(i) := \left(\sum_{i=a}^{b} f(i)\right) + f(\mathfrak{suc}(b)) \quad \text{where } b \in \mathbb{N} \text{ such that } a \leq b$$

For completeness, we say $\sum_{i=a}^{b} f(i) := 0$ when $b \in \mathbb{N}$ such that $b < a$. For this problem, you may also assume associativity and commutativity of multiplication over $\mathbb{N}$.

Prove that $(\forall n \in \mathbb{N})\left(1 + \sum_{i=0}^{n} 2^i = 2^{n+1}\right)$.

Proof. We will prove $(\forall n \in \mathbb{N})\left(1 + \sum_{i=0}^{n} 2^i = 2^{n+1}\right)$ by weak induction.

Basis Step: We observe the following. In the basis step, we need to prove $1 + \sum_{i=0}^{0} 2^i = 2^{0+1}$.

$$\begin{aligned}
1 + \sum_{i=0}^{0} 2^i &= 1 + 2^0 && \text{by definition of iterated summation} \\
&= 1 + 1 && \text{by definition of exponentiation} \\
&= 2 && \text{by definition of } + \\
&= 2^1 && \text{since } 2^1 = 2^{\mathfrak{suc}(0)} = 2^0 \cdot 2 = 1 \cdot 2 = 2 \\
&= 2^{0+1} && \text{since } 0 + 1 = 1
\end{aligned}$$

Therefore $1 + \sum_{i=0}^{0} 2^i = 2^{0+1}$ as desired.

Inductive Step: Let $k \in \mathbb{N}$, and assume $1 + \sum_{i=0}^{k} 2^i = 2^{k+1}$. In the inductive step, we must prove $\mathbf{1} + \sum_{i=0}^{\mathfrak{suc}(k)} \mathbf{2}^i = \mathbf{2}^{\mathfrak{suc}(k)+1}$. We observe the following.

$$
\begin{aligned}
\mathbf{1} + \sum_{i=0}^{\mathfrak{suc}(k)} \mathbf{2}^i &= \mathbf{1} + \left(\sum_{i=0}^{k} \mathbf{2}^i\right) + \mathbf{2}^{\mathfrak{suc}(k)} && \text{by definition of iterated summation} \\
&= \left(\mathbf{1} + \sum_{i=0}^{k} \mathbf{2}^i\right) + \mathbf{2}^{\mathfrak{suc}(k)} && \text{by associativity of } + \\
&= \mathbf{2}^{k+1} + \mathbf{2}^{\mathfrak{suc}(k)} && \text{by the inductive hypothesis} \\
&= \mathbf{2}^{k+1} + \mathbf{2}^{k+1} && \text{since } (\forall x \in \mathbb{N})(\mathfrak{suc}(x) = x + \mathbf{1}) \\
&= \mathbf{2} \cdot \mathbf{2}^{k+1} && \text{by commutativity of } \cdot \text{ and definition of } \cdot \\
&= \mathbf{2}^{k+1} \cdot \mathbf{2} && \text{by commutativity of } \cdot \\
&= \mathbf{2}^{\mathfrak{suc}(k+1)} && \text{by definition of exponentiation} \\
&= \mathbf{2}^{(k+1)+1} && \text{since } (\forall x \in \mathbb{N})(\mathfrak{suc}(x) = x + \mathbf{1}) \\
&= \mathbf{2}^{\mathfrak{suc}(k)+1} && \text{by associativity of } +
\end{aligned}
$$

Thus, we have shown $\mathbf{1} + \sum_{i=0}^{\mathfrak{suc}(k)} \mathbf{2}^i = \mathbf{2}^{\mathfrak{suc}(k)+1}$ as required.

Therefore, we can conclude $(\forall n \in \mathbb{N})\left(\mathbf{1} + \sum_{i=0}^{n} \mathbf{2}^i = \mathbf{2}^{n+1}\right)$.